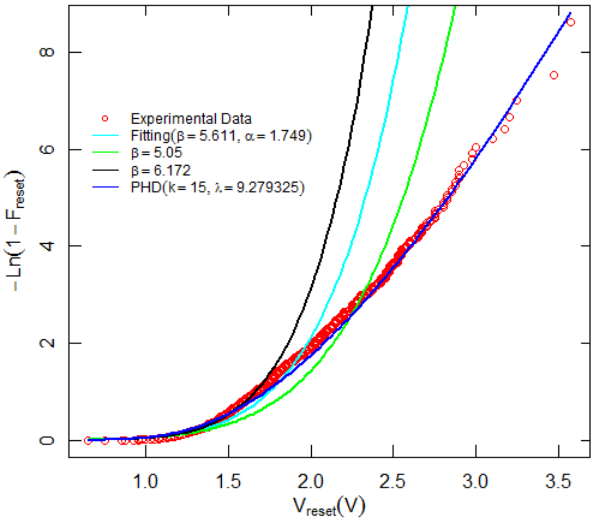

Figure 3. Cumulative hazard rate of V_{reset} for 2749 RS cycles and the corresponding Weibull and PH distribution fit.

The best result is achieved when the Erlang distribution is considered and the accuracy of the fit is remarkable, as shown in Figure 3.

Once the statistical analysis has been developed, a detailed study of the devices considered here is developed. To do so, an analysis based on the data screened by means of the Low Resistance State (LRS) of the device resistance, R, is performed. The device LRS resistance is measured just after a set process is over, when the conductive filament is fully formed. Usually, the resistance at this point is the lowest value found all along a complete resistive switching cycle.

If V_{reset} values are sorted out by considering the LRS resistance, a better fit is obtained by means of WD, although the fitting is not accurate, as can be seen in Figure 4a. In particular, for R<20 kΩ the fitting is not very good. Nevertheless, if a PHD is employed instead of a WD a reasonable accuracy is achieved. In this respect, the PHD appropriateness to deal with our experimental data is noteworthy at the sight of Figure 4b, and this is for all the resistance range under consideration.

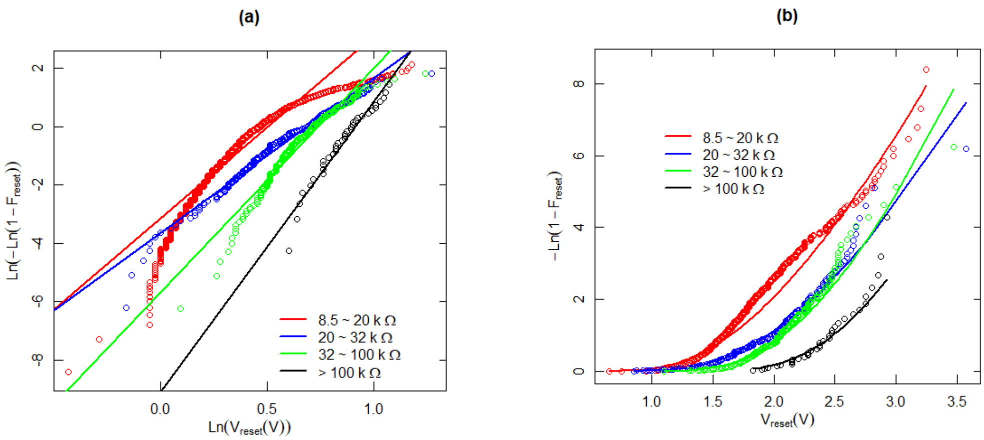

Figure 4. a) WD Weibits versus Ln(V_{reset}) for the experimental data under consideration screened for different LRS resistances are plotted in symbols. The analytical WD best fit is also shown in solid lines, b) hazard rate for the screened experimental data (symbols) and PHD (solid lines).

The reliability function, i.e., the survival function, as it is known in scientific branches not related to engineering, is interesting to analyze the statistical properties of the data we are dealing with. Since V_{reset} can be considered as the failure voltage for the memories under study (RRAMs), the reliability functions portrays the probability that a memory state change will not be produced; i.e., the conductive filament will not be broken and the memory resistance state will not be switched. From another viewpoint, the reliability function describes the probability that the conductive filament will not be ruptured for voltages below the failure voltage.

The reliability function has been plotted in Figure 5 for the experimental data, WDs and PHDs. Although no distribution shows a close reproduction of the experimental values for low voltages, at medium-high voltages the PHD works better than WD and achieves a reasonably good performance.

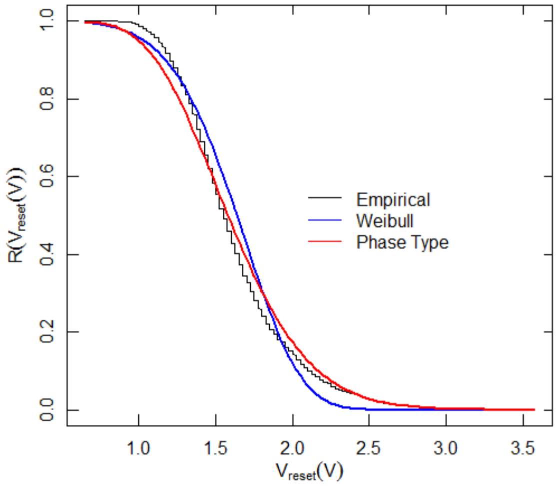

Figure 5. Reliability function versus V_{reset} for experimental data and Weibull and Phase-type distributions.

In order to further characterize the correctness of our approach, the hazard rate function should also be considered since it describes the failure rate in a voltage interval (V_{reset},V_{reset}+dV_{reset}). It could also be interpreted as the device degradation velocity at a certain voltage. This function has been plotted in Figure 6. Again, as can be seen and it was expected from previous results, PHD works better than WD.

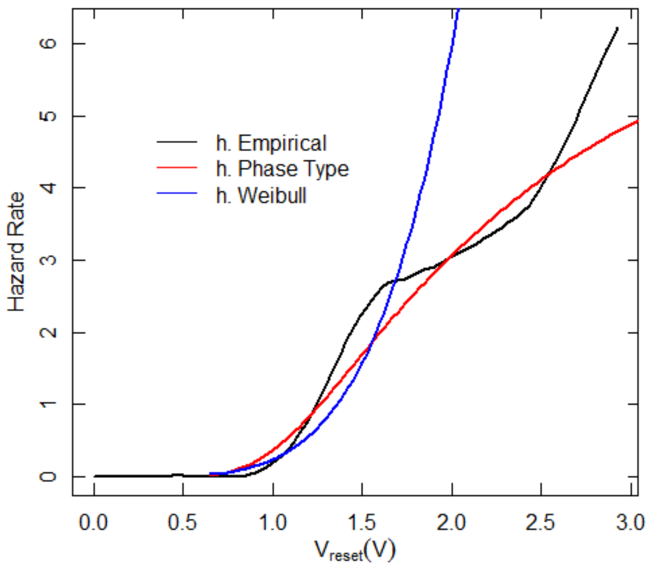

Figure 6. Hazard rate versus V_{reset} for experimental data, phase-type and Weibull distributions.